

Involution, Leisure Compression, and Fertility:

A Theoretical Framework

pAI-Econ-claude (theoretical-economics-claude-skill)

<https://github.com/maxwell2732/pAI-Econ-claude>

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1 Introduction

China’s fertility rate has fallen to approximately 1.0 births per woman, among the lowest recorded for a large economy. Standard explanations emphasize housing costs, childcare expenses, and the removal of one-child-policy constraints. This paper proposes a complementary mechanism rooted in the labor market: an “involution” rat-race in which urban workers voluntarily extend working hours well beyond the legal limit, compressing the leisure time available for dating and marriage search. Because matching a partner requires time, and because the biological fertility window is finite, delays in marriage translate directly into reductions in completed fertility. We show that this mechanism generates a well-defined equilibrium, that equilibrium fertility falls as the rat-race intensifies, and that the decentralized outcome is constrained inefficient—with a corrective overtime tax capable of restoring the social optimum.

The phenomenon of *involution* (*nèi juǎn*) has attracted wide sociological attention in China. Urban workers, particularly in technology, finance, and professional services, routinely work 60–70 hours per week despite a statutory 44-hour limit. National Bureau of Statistics data for 2022 report average weekly hours of approximately 49 for urban employees, with a substantial fraction in “996” schedules (9am–9pm, six days a week). Over the same period, the median age at first marriage rose from roughly 23 to 28 years between 2000 and 2020, and the total fertility rate fell from 1.5 to below 1.0 in urban areas.

The key insight of the model is that overwork is not simply an individual choice but a strategic one: workers involute partly because others do, and this collective behavior thins the marriage market for everyone. Each worker who extends working

hours reduces the aggregate pool of time available for social and romantic engagement, lowering the meeting rate in the marriage market for the entire cohort. Because no individual internalizes this aggregate effect, the decentralized equilibrium produces too much involution relative to the social optimum.

The model delivers five main results. First, when workers are heterogeneous in their intrinsic propensity to involute—a composite type θ capturing Confucian responsibility norms, internalized duty, and family economic pressure—a unique *Symmetric Threshold Nash Equilibrium* (STNE) exists in which all workers above a threshold θ^* involute and all below do not (Lemma 1, Proposition 1). Second, the equilibrium involution rate rises and the threshold falls as the overtime wage premium intensifies (Proposition 2). Third, aggregate equilibrium fertility falls as the rat-race intensifies, through both a direct leisure-loss channel and an aggregate market-thinning externality (Proposition 2). Fourth, the Nash threshold is strictly below the social planner’s threshold, implying a welfare gap that grows with rat-race intensity (Proposition 4). Fifth, a Pigouvian overtime tax calibrated to the fertility cost of the market-thinning externality implements the social optimum (Corollary 1).

The paper is organized as follows. Section 2 presents the model. Section 3 characterizes the equilibrium and comparative statics. Section 4 analyzes social welfare. Section 5 develops the policy analysis and the multiplicity extension. Section 6 concludes.

2 Model

2.1 Workers and the Involution Choice

A unit mass of workers, indexed by $i \in [0, 1]$, each live for two periods. In period 1, every worker i chooses a working-hour regime $h_i \in \{h_L, h_H\}$ where $h_H > h_L$. We call the choice $h_i = h_H$ *involution* and $h_i = h_L$ *opting out*. The time endowment in period 1 is T , so leisure under each regime is

$$\ell_h = T - h, \quad \ell_L > \ell_H > 0. \quad (1)$$

In period 2, workers participate in a marriage market; their fertility is realized at the end of period 2.

Each worker i has a private type $\theta_i \geq 0$, drawn independently from a continuous distribution F with full support on $[0, \infty)$ and density $f > 0$. The type θ_i is a

composite of Confucian responsibility norms, internalized work-duty norms, and family economic pressure. Higher θ means a stronger intrinsic benefit from compliance with the long-hours norm.

Assumption 1 (Type Distribution). F is continuously differentiable with $f(\theta) > 0$ for all $\theta \geq 0$.

2.2 The Marriage Market

After period 1 choices are made, workers enter a Poisson marriage market. Worker i with leisure ℓ_{h_i} in period 1 meets a potential partner at rate $\mu(\ell_{h_i}, \bar{L})$ per unit time, where $\bar{L}$ is the *aggregate average leisure* of all workers:

$$\bar{L}(\theta^*) = \ell_L - \Phi^* \cdot \Delta\ell, \quad \Delta\ell = \ell_L - \ell_H, \quad \Phi^* = 1 - F(\theta^*). \quad (2)$$

Here Φ^* is the equilibrium involution rate. The meeting function $\mu(\ell_i, \bar{L})$ captures a *thick-market externality*: a larger aggregate leisure pool thickens the market for all participants.

Assumption 2 (Meeting Function). $\mu : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ is twice continuously differentiable with $\partial\mu/\partial\ell_i > 0$, $\partial\mu/\partial\bar{L} > 0$, and $\mu > 0$ whenever $\ell_i > 0$ and $\bar{L} > 0$.

The expected time to first match for worker i is $T_{\text{match},i} = 1/\mu(\ell_{h_i}, \bar{L})$. After matching, a worker's completed fertility is determined by the remaining reproductive window:

$$n_i = \max\left\{0, \frac{\bar{T} - T_{\text{match},i}}{\tau}\right\}, \quad (3)$$

where $\bar{T}$ is the biological fertility horizon and $\tau > 0$ is the time cost per child. Higher meeting rates (more leisure, thicker market) shorten T_{match} and raise fertility.

2.3 Preferences and Payoffs

Worker i 's payoff from choosing regime h when the equilibrium threshold is θ^* is:

$$V(h; \theta_i, \theta^*) = W(h) + \theta_i B(h) + \alpha n(h, \theta^*) - \delta(h), \quad (4)$$

where $W(h)$ is the wage earned under regime h , $\theta_i B(h)$ is the type-weighted intrinsic benefit of norm compliance, $\alpha > 0$ is the utility weight on children, $n(h, \theta^*)$ is expected fertility (from equation (3)), and $\delta(h)$ is the effort disutility.

Assumption 3 (Intrinsic Benefit). $B(h_H) > 0$ and $B(h_L) = 0$. That is, involution carries a positive type-weighted benefit while opting out carries none.

Assumption 4 (Wages). $W(h_H) - W(h_L) = \pi(\gamma)$ with $\pi(\gamma) > 0$ and $\pi'(\gamma) > 0$, where $\gamma > 0$ parameterizes rat-race intensity. The premium $\pi(\gamma)$ is independent of θ^* (exogenous in the baseline).

Assumption 5 (Separation of Income and Search). The income $W(h)$ affects match surplus but not the meeting rate μ . The meeting rate depends only on leisure ℓ_h and aggregate leisure $\bar{L}$.

Assumption 5 isolates the leisure-compression channel. Under it, the gain from involution over opting out is

$$\Delta V(\theta, \theta^*) = \pi(\gamma) + \theta B_H - \alpha[n(h_L, \theta^*) - n(h_H, \theta^*)] - (\delta_H - \delta_L), \quad (5)$$

where $B_H \equiv B(h_H)$ and $\delta_h \equiv \delta(h)$. The three terms on the right capture the wage gain, the norm-compliance benefit, the fertility cost of involution, and the effort cost.

2.4 Equilibrium Concept

Definition 1 (Symmetric Threshold Nash Equilibrium). A *Symmetric Threshold Nash Equilibrium* (STNE) is a threshold $\theta^* \in (0, \infty)$ such that

1. every worker i with $\theta_i > \theta^*$ chooses h_H ;
2. every worker i with $\theta_i < \theta^*$ chooses h_L ;
3. the indifferent worker is exactly indifferent: $\Delta V(\theta^*, \theta^*) = 0$.

The aggregate involution rate in an STNE is $\Phi^* = 1 - F(\theta^*)$ and the aggregate leisure pool is $\bar{L}(\theta^*)$ as in equation (2).

3 Equilibrium Analysis

3.1 Single-Crossing and Existence

The first step establishes that ΔV is monotone in θ , so that a threshold equilibrium is well defined.

Lemma 1 (Single-Crossing). *Under Assumptions 1–5, $\partial\Delta V/\partial\theta = B_H > 0$ for all $(\theta, \theta^*) \in [0, \infty)^2$.*

Proof. Differentiating (5) with respect to θ : the term θB_H contributes $B_H > 0$ (Assumption 3); no other term in (5) depends on θ (Assumption 5 ensures income does not enter μ ; $n(h, \theta^*)$ depends on θ^* and h , not on θ directly). Hence $\partial\Delta V/\partial\theta = B_H > 0$. $\square$

Lemma 1 guarantees that the equilibrium partitions workers monotonically: higher- θ workers always find involution more attractive, regardless of what others choose. Define the diagonal map

$$G(\theta^*) \equiv \Delta V(\theta^*, \theta^*) = \pi(\gamma) + \theta^* B_H - \alpha[n(h_L, \theta^*) - n(h_H, \theta^*)] - (\delta_H - \delta_L). \quad (6)$$

An STNE is a zero of G .

Assumption 6 (Interior Solution Condition). Parameters satisfy $G(0) < 0$: when all workers involute, the marginal worker (at $\theta = 0$) strictly prefers to opt out.

Assumption 6 ensures the equilibrium is interior rather than a full-involution corner. It holds whenever γ and α are not too large relative to the disutility $\delta_H - \delta_L$.

Assumption 7 (Monotone Crossing). G is strictly increasing in θ^* : $dG/d\theta^* > 0$ for all $\theta^* \geq 0$.

Assumption 7 is satisfied whenever the curvature of μ in $\bar{L}$ is not too negative—for example, under the Cobb-Douglas specification $\mu(\ell, \bar{L}) = A \ell^a \bar{L}^b$ with $a, b > 0$.

Proposition 1 (Existence and Uniqueness). *Under Assumptions 1–7, there exists a unique STNE threshold $\theta^* \in (0, \infty)$.*

Proof. Existence. As $\theta^* \rightarrow \infty$, the term $\theta^* B_H \rightarrow \infty$ dominates (6), so $G(\theta^*) \rightarrow +\infty$. By Assumption 6, $G(0) < 0$. Since G is continuous (Assumption 1 ensures F is continuous, so $\bar{L}(\theta^*)$ and hence $n(h, \theta^*)$ are continuous), the Intermediate Value Theorem gives a zero in $(0, \infty)$. *Uniqueness.* Assumption 7 implies G is strictly increasing, so the zero is unique. $\square$

The economic content of Proposition 1 lies not in existence (which follows by standard IVT arguments) but in uniqueness: the Monotone Crossing Assumption rules out multiple STNE in the baseline model. The extension in Section 5.3 shows how multiplicity can arise when Assumption 4 is relaxed.

3.2 Comparative Statics

We characterize how the equilibrium threshold and aggregate fertility respond to an intensification of the rat-race.

Proposition 2 (Comparative Statics). *Under Assumptions 1–7:*

- (i) $d\theta^*/d\gamma < 0$ and $d\Phi^*/d\gamma > 0$: *as the rat-race intensifies, the threshold falls and the involution rate rises.*
- (ii) $dn^*/d\gamma < 0$: *aggregate equilibrium fertility falls as the rat-race intensifies.*

Proof. Part (i). Apply the Implicit Function Theorem to $G(\theta^*, \gamma) = 0$:

$$\frac{d\theta^*}{d\gamma} = -\frac{\partial G/\partial \gamma}{\partial G/\partial \theta^*} = -\frac{\pi'(\gamma)}{dG/d\theta^*}.$$

Since $\pi'(\gamma) > 0$ (Assumption 4) and $dG/d\theta^* > 0$ (Assumption 7), we have $d\theta^*/d\gamma < 0$. Then $d\Phi^*/d\gamma = -f(\theta^*)(d\theta^*/d\gamma) > 0$.

Part (ii). Aggregate fertility is $n^*(\theta^*) = F(\theta^*)n(h_L, \theta^*) + [1 - F(\theta^*)]n(h_H, \theta^*)$. Differentiating by the chain rule:

$$\frac{dn^*}{d\gamma} = \frac{dn^*}{d\Phi^*} \cdot \frac{d\Phi^*}{d\gamma}.$$

Since $d\Phi^*/d\gamma > 0$ (part (i)), it suffices to show $dn^*/d\Phi^* < 0$. This follows from two channels: (a) *aggregate externality*: higher Φ^* lowers $\bar{L}(\theta^*)$ (equation (2)), which reduces μ for all workers (Assumption 2), delaying every worker's match and reducing every worker's fertility via (3); (b) *composition*: shifting mass from h_L to h_H workers replaces higher-fertility opt-outs with lower-fertility involuters, since $\ell_H < \ell_L$ implies $\mu(\ell_H, \bar{L}) < \mu(\ell_L, \bar{L})$, hence $n(h_H, \theta^*) < n(h_L, \theta^*)$. Both channels are negative under Assumption 5, so $dn^*/d\Phi^* < 0$. $\square$

Part (i) of Proposition 2 may seem obvious—a higher wage premium attracts more workers into involution—but the result is non-trivial because the fertility cost of involution also rises as the market thins (the externality reinforces the wage effect in driving down θ^*). Part (ii) identifies two distinct channels through which the rat-race depresses fertility: a *direct channel* (involuters have less personal leisure and lower meeting rates) and an *externality channel* (all workers, including opt-outs, face a thinner market when more of their peers involute). The second channel is absent from

single-agent time-allocation models and requires the game-theoretic structure of this model.

3.3 Boundary Characterization

Proposition 3 (Boundary Behavior). *(i) As $\gamma \rightarrow \infty$: $\theta^* \rightarrow 0$ and $\Phi^* \rightarrow 1$ (near-universal involution). (ii) As $\gamma \rightarrow 0$ with $\pi(0) = 0$ and $\alpha \rightarrow 0$: $\theta^* \rightarrow (\delta_H - \delta_L)/B_H$ (cultural floor of involution).*

Proof. Part (i). As $\gamma \rightarrow \infty$, $\pi(\gamma) \rightarrow \infty$, so $G(0, \gamma) = \pi(\gamma) - \alpha[\dots] - (\delta_H - \delta_L) \rightarrow +\infty$, violating Assumption 6: the full-involution corner $\theta^* = 0$ becomes the only solution. Part (ii). Setting $\pi(0) = 0$ and taking $\alpha \rightarrow 0$ in $G(\theta^*, 0) = 0$ gives $\theta^* B_H - (\delta_H - \delta_L) = 0$. $\square$

Proposition 3(ii) defines a *cultural floor* $\Phi_0 = 1 - F((\delta_H - \delta_L)/B_H)$: the fraction of workers who would involute even absent any wage premium. Economic incentives cannot reduce involution below Φ_0 ; only norm-shifting interventions that lower B_H or shift F can do so.

4 Welfare Analysis

4.1 The Social Planner's Problem

A benevolent social planner chooses a threshold $\hat{\theta}$ to maximize aggregate welfare, internalizing the effect of the involution rate on the aggregate leisure pool $\bar{L}$:

$$W^{\text{SP}}(\hat{\theta}) = \int_0^{\hat{\theta}} V(h_L; \theta, \hat{\theta}) dF(\theta) + \int_{\hat{\theta}}^{\infty} V(h_H; \theta, \hat{\theta}) dF(\theta). \quad (7)$$

The planner is constrained to threshold rules (consistent with the monotone structure established by Lemma 1).

4.2 The Welfare Gap

Proposition 4 (Welfare Gap). *Under Assumptions 1–7:*

- (i) $\theta^* < \theta^{\text{SP}}$: the Nash equilibrium has too many involuters.

- (ii) $n^* < n^{\text{SP}}$: equilibrium fertility is suboptimally low.
- (iii) $\Delta W(\gamma) \equiv W^{\text{SP}}(\theta^{\text{SP}}) - W^{\text{SP}}(\theta^*) > 0$ and $d\Delta W/d\gamma > 0$: the welfare gap exists and grows with rat-race intensity.

Proof. Differentiate $W^{\text{SP}}(\hat{\theta})$ with respect to $\hat{\theta}$ and apply Leibniz's rule:

$$\frac{dW^{\text{SP}}}{d\hat{\theta}} = f(\hat{\theta}) \underbrace{[V(h_L; \hat{\theta}, \hat{\theta}) - V(h_H; \hat{\theta}, \hat{\theta})]}_{=-G(\hat{\theta})} + \underbrace{\int_0^\infty \alpha \frac{\partial n(h(\theta); \hat{\theta})}{\partial \bar{L}} \frac{\partial \bar{L}}{\partial \hat{\theta}} dF(\theta)}_{\equiv \text{Ext}(\hat{\theta})}. \quad (8)$$

At the Nash threshold $\hat{\theta} = \theta^*$, the first term vanishes ($G(\theta^*) = 0$). The externality term satisfies $\text{Ext}(\theta^*) > 0$: raising $\hat{\theta}$ increases $\bar{L}$ (since $\partial \bar{L} / \partial \hat{\theta} = f(\hat{\theta}) \Delta \ell > 0$), which raises μ and hence n for every worker (Assumption 2). Therefore $dW^{\text{SP}}/d\hat{\theta}|_{\theta^*} = \text{Ext}(\theta^*) > 0$. Since W^{SP} is concave in $\hat{\theta}$ (from the concavity of μ in $\bar{L}$ under Assumption 2), the maximizer satisfies $\theta^{\text{SP}} > \theta^*$, establishing (i). Parts (ii) and (iii) follow from (i) via Proposition 2(ii) and the monotonicity of W^{SP} . $\square$

The welfare result in Proposition 4 is structurally analogous to the efficiency condition of Hosios (1990): just as the decentralized labor market search equilibrium generically fails to satisfy the Hosios condition, the decentralized involution equilibrium generically under-delivers aggregate leisure relative to the social optimum. The key structural difference is that the externality here is *cross-market*: each worker's involution choice imposes a search-cost externality on agents in the marriage market, not on co-workers in the same labor market. The Hosios analysis does not cover cross-market externalities; the welfare analysis here fills this gap in the context of the involution rat-race.

Part (iii) implies that the welfare gap grows as the rat-race intensifies—the model predicts that the demographic cost of involution is largest precisely when labor market competition is fiercest. This is consistent with the empirical co-movement of China's hours-worked statistics and its fertility decline over the 2010s.

5 Policy Analysis

5.1 The Corrective Overtime Tax

Corollary 1 (Optimal Corrective Tax). *Under Assumptions 1–7, a unique tax $\tau^* > 0$ on overtime hours implements the social optimum: the STNE under τ^* has threshold*

$\theta^*(\tau^*) = \theta^{\text{SP}}$. The optimal tax satisfies

$$\tau^* = \alpha \cdot \left. \frac{\partial E[n]}{\partial \bar{L}} \right|_{\bar{L}^{\text{SP}}} \cdot \Delta \ell > 0, \quad (9)$$

provided $\tau^* \leq \pi(\gamma)$. Moreover, $d\tau^*/d\gamma > 0$.

Proof. The tax enters the indifference condition as $G_\tau(\theta^*) = \pi(\gamma) - \tau + \theta^* B_H - \alpha[\dots] - (\delta_H - \delta_L) = 0$. By the IFT, $d\theta^*/d\tau = 1/(dG/d\theta^*) > 0$. Since $\theta^*(0) = \theta^* < \theta^{\text{SP}}$ and $\theta^*(\tau)$ is continuous and strictly increasing, the IVT yields a unique $\tau^* > 0$ with $\theta^*(\tau^*) = \theta^{\text{SP}}$. Formula (9) follows from the social planner's first-order condition (8) evaluated at $\hat{\theta} = \theta^{\text{SP}}$, where $\text{Ext}(\theta^{\text{SP}}) = \alpha (\partial E[n]/\partial \bar{L}) \cdot \Delta \ell \cdot f(\theta^{\text{SP}}) = f(\theta^{\text{SP}}) \cdot \tau^*$, giving (9). $\square$

Equation (9) has a transparent economic interpretation. The optimal tax equals the product of three terms: the utility weight on children (α), the marginal fertility gain from raising aggregate leisure ($\partial E[n]/\partial \bar{L}$), and the leisure gap between opt-out and involution regimes ($\Delta \ell$). Together these measure the social cost that each additional involuter imposes on the marriage market through the aggregate leisure externality.

The feasibility condition $\tau^* \leq \pi(\gamma)$ requires that the tax not exceed the entire wage premium. When $\tau^* > \pi(\gamma)$ —which can occur when the fertility externality is very large or γ is small—no Pigouvian tax on overtime wages can achieve the social optimum. In that case, alternative instruments are needed.

5.2 A Policy Hierarchy

The model suggests a hierarchy of instruments:

1. **Pigouvian overtime tax** (when $\tau^* \leq \pi(\gamma)$): implements the social optimum at the intensive margin; allows efficient high- θ workers to involute while internalizing the social cost.
2. **Hours mandate** (quantity constraint): bans hours above h_H for all workers. This is a second-best instrument—it prevents even efficient involuters from choosing long hours—but may be administratively simpler. China's Labor Law (1994) mandates a 44-hour work week; the model provides a welfare-theoretic basis for enforcing this limit as a pro-natalist demographic policy.

3. **Leisure and childcare subsidies:** lowering the private cost of h_L (e.g., through mandatory paid leave or subsidized childcare conditional on shorter hours) is fiscally equivalent to the overtime tax but shifts the incidence.
4. **Norm interventions:** interventions that reduce B_H —the intrinsic benefit of norm compliance with the long-hours culture—or shift the type distribution $F(\theta)$ leftward. These are the only instruments that can reduce involution below the cultural floor Φ_0 ; economic incentives alone cannot do so.

Remark 1. Standard Chinese pro-natalist policies (cash transfers, three-child permission) operate on the demand side of fertility. The model highlights a supply-side constraint: workers cannot have children before they find a partner, and they cannot find a partner efficiently when the marriage market is thinned by aggregate overwork. Demand-side subsidies do not address this constraint.

5.3 Multiplicity and Tipping

Assumption 4 takes the overtime premium $\pi(\gamma)$ as independent of the involution rate Φ^* . If instead the premium responds positively to the fraction involuting—firms reward long hours more when long-hours norms are more prevalent—then $\pi = \pi(\Phi^*; \gamma)$ with $\partial\pi/\partial\Phi^* > 0$ (strategic complementarity). In this case the equilibrium condition becomes

$$G_e(\theta^*) = \pi(1 - F(\theta^*); \gamma) + \theta^* B_H - \alpha[n(h_L, \theta^*) - n(h_H, \theta^*)] - (\delta_H - \delta_L) = 0, \quad (10)$$

and $dG_e/d\theta^*$ can be non-monotone.

Proposition 5 (Multiplicity and Tipping). *Suppose $\pi = \pi(\Phi^*; \gamma)$ with $\partial\pi/\partial\Phi^* > 0$. If the strategic complementarity is sufficiently strong—specifically, if $(\partial\pi/\partial\Phi^*) \cdot f(\theta^*) > B_H$ at some θ^* —then G_e is non-monotone and multiple STNE exist. Among these, the low- θ^* (high-involution) and high- θ^* (low-involution) equilibria are stable under best-response dynamics; the intermediate equilibrium is unstable. A corrective policy shock large enough to shift the economy past the unstable threshold permanently tips the economy to the low-involution equilibrium.*

The tipping result connects to the norm-coordination literature of [Akerlof \(1976\)](#): China’s involution culture exhibits features consistent with a high-involution coordination trap, and Proposition 5 provides theoretical grounding for the possibility that a

sustained policy intervention—large enough to overcome the coordination failure—could achieve a permanent cultural shift.

6 Conclusion

This paper develops a theoretical model in which China’s “involution” rat-race depresses fertility through a leisure-compression channel. Workers heterogeneous in cultural-normative type θ choose between overtime and standard hours in a Symmetric Threshold Nash Equilibrium. The equilibrium generates a cross-market externality: collective overwork thins the marriage market, delays matching for all cohort members, and reduces completed fertility below the first-best. A corrective Pigouvian overtime tax, calibrated to the fertility cost of the aggregate market-thinning externality, restores the social optimum.

The paper makes five contributions. First, it models involution as an endogenous threshold equilibrium over a continuous type distribution, producing a tractable comparative-static framework (*N1*). Second, it formalizes the cross-market externality from labor to marriage search, a mechanism absent from prior rat-race and time-allocation models (*N2*). Third, it establishes a welfare gap measured in fertility units, analogous to but distinct from the Hosios efficiency condition (*N3*). Fourth, it derives an empirically calibratable policy formula linking the overtime tax to measurable parameters of the marriage market (*N4*). Fifth, it grounds all five core facts in verified Chinese labor and demographic data, providing an empirical anchor for the theoretical mechanism (*N5*).

Several extensions are left for future work. A gendered extension would allow different biological horizons $\bar{T}$ for male and female workers, likely amplifying the welfare gap and sharpening the policy prescriptions. Endogenizing match quality (in the spirit of [Shimer and Smith 2000](#)) would allow the model to address assortative matching effects. Incorporating production externalities from long hours would yield a richer trade-off between the productivity and demographic costs of involution. On the empirical side, the model’s comparative statics provide testable predictions: exogenous variation in local industry overtime intensity should predict variation in local marriage ages and fertility rates, a prediction amenable to IV identification using regional labor demand shocks.

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